
Amazon CloudFront – Caching & Geographic Features

What is Amazon CloudFront?

Amazon CloudFront is a **Content Delivery Network (CDN)** that delivers your content (websites, APIs, videos, software) to users globally through a **network of edge locations**. It ensures **low latency, high transfer speeds, and security**.

Part 1: CloudFront Caching – How It Works

1. Caching Layer Overview

- CloudFront caches content at **AWS Edge Locations** (PoPs) around the globe.
 - When a user requests content:
 - CloudFront checks the **cache** at the nearest Edge.
 - If the object is **cached**, it is returned ( **cache hit**).
 - If not, CloudFront fetches it from the **origin** (S3, EC2, ALB, or external server), caches it, then returns it ( **cache miss**).
-

2. What Can Be Cached?

- **Static content:** images, CSS, JS, HTML
- **Dynamic content:** API responses, custom headers (with proper settings)

- **Signed URLs** or **Signed Cookies** for private content

3. TTL (Time To Live) Settings

Type	Meaning	Default
Minimum TTL	Shortest time content stays in cache	0 seconds
Maximum TTL	Longest time content can stay in cache	31536000 seconds (1 year)
Default TTL	Default caching time	86400 seconds (1 day)

 Use these to balance **performance vs freshness** of data.

4. Cache Invalidation

- You can force CloudFront to remove cached content before TTL expires.
- Useful when updating website content (e.g., new image version).
- Invalidation via:
 - AWS Console

AWS CLI:

```
aws cloudfront create-invalidation --distribution-id ABC123 --paths "/*"
```

-

5. Cache Behavior Customization

- Based on:
 - **Headers** (e.g., **User-Agent**)
 - **Cookies**
 - **Query Strings**
 - You can configure which values CloudFront should forward to the origin and vary the cache key by.
-

Part 2: Geographic Features

1. Edge Locations

- Global data centers where content is cached.
 - Users are routed to the **nearest edge location** using **AWS Route 53**.
-

2. Geo Targeting (Geo Restriction / Geo Blocking)

Feature	Description
Geo Restriction	Block access to content from certain countries
Use Cases	Legal restrictions, licensing, content compliance

 Configure via CloudFront **Geo Restriction settings**.

3. Geographic Routing & Behavior

- You can **serve different content** based on user's region using:
 - **Lambda@Edge functions**
 - **CloudFront Functions** (lighter, faster)
- Examples:
 - Show different languages
 - Redirect users to regional pages

4. Edge Authorization (Geo-aware Security)

- You can block regions at **WAF layer** or via CloudFront **Geo Restriction**.
- Add IP-level restrictions using AWS WAF with CloudFront.

Pricing Consideration

- Pricing depends on:
 - **Data transfer out** (varies by geographic region)
 - **Number of requests**

- **Invalidation requests**
 - **Field-level encryption or Lambda@Edge usage**
-

Teaching Tip: Demo Ideas

- Serve a static website via CloudFront and simulate:
 - TTL behavior (modify content, observe delay)
 - Invalidation
 - Geo restriction (use VPN or curl with IP spoofing)
 - Viewer location detection using CloudFront Function
-